

Surviving extremes

Some animals thrive in almost unbelievable extremes. In volcanic vents on the seabed, Pompeii worms endure water temperatures of up to 176°F (80°C), while in the Arctic, wood frogs and insects survive being frozen solid for several months each year. In the highest mountains, birds routinely feed at heights of 26,000 ft (8,000 m), where a human would be left gasping for breath. In many habitats, every year brings its share of heat, cold, or drought, and animals often survive by shutting down when life gets tough. When conditions improve, they become active once more.



Heat freak

Scientists discovered Pompeii worms in the 1980s when exploring superhot volcanic vents on the Pacific Ocean floor. The vent water can reach 750°F (400°C), but the worms live in a narrow “comfort zone,” between the vents and the surrounding cold ocean. They keep their heads in cold water, while their tails take the heat.

Hibernation

In winter, cold weather often makes food hard to find. Many mammals, including the common dormouse, survive by going into hibernation. True hibernation is very different from sleep. When an animal hibernates, its body temperature drops to a very low level, and its heart beats only a few times each minute. It gets its energy from reserves of body fat, which it stores up before hibernation starts.